

UME Seminar

Date and Time: June 5, 2026 (11 - 12 am)

Venue: Amphi 104 (Pôle Méca)

Close look at jumping microdrops

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Abstract

On most solid surfaces, two merging drops form a single larger one that sticks to the substrate. In contrast, coalescence can trigger droplet jumping on a surface composed of a dense network of hydrophobic nanocones. Mouterde et al. showed that the probability of ejection of micro droplets can then be as high as 99%, meaning that such materials have very unusual anti-fogging abilities. We first discuss what happens after the drop departure. Two post-coalescence scenarios are identified. %
1) For small drops ($R < 10 \mu\text{m}$), 90% of ejections leave the solid dry. 2) For large drops ($R > 33 \mu\text{m}$), the solid after the liquid departure is coated by many droplets that induce multiple nucleation events. We try to understand these differences. In addition, we show more generally that the nucleation of drops on these cold anti-fogging materials is attenuated. It seems to happen mainly on “mobile hydrophilic defects” such as dust, so that drop departures also contribute to surface cleaning, besides its anti-dew function. Lastly, we decrease surface temperature down to -10°C , at which droplets may solidify. However, we observe that they still jump, demonstrating an anti-icing effect, since that drops are ejected before they have time to freeze.

Predicting the statistics of near-inertial waves over a background flow via statistical mechanics

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Abstract

In the Ocean, Earth’s rotation is responsible for Near-Inertial Waves (NIWs) oscillating near the Coriolis frequency. These waves account for a significant fraction of the Ocean’s kinetic energy, and can induce turbulent mixing at depth. Understanding their dynamics is thus crucial to parameterize their impact on the climate system. In this presentation, I will focus on the effects of oceanic currents and eddies: the NIW field is forced at large scales by atmospheric storms, and subsequently develops smaller scales through advection and refraction. I will discuss how their statistics can be computed using standard expansions and tools borrowed from statistical mechanics: spatial distributions of wave kinetic energy, of wave potential energy, and of the Stokes drift induced by the waves. The predictions are quantitatively validated against numerical simulations. This approach can be generalized to a wide class of wave systems in inhomogeneous media with currents, such as gravity waves over topography or capillary waves, among others.